Neural DNA: A Compact Genome for Growing Network Architecture

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Abstract

Neural network architecture design requires expensive human expertise or costly automated search. We introduce Neural DNA (NDNA), a compact learned genome of fewer than 400 parameters that grows network topology through type-based compatibility rules, default-disconnected initialization, and metabolic cost pressure. The genome assigns cell types to neurons based on their position, computes pairwise connection probabilities from type compatibility, and applies the resulting binary masks to network weights. We evaluate NDNA across four architectures (MLP, CNN, Transformer, Video Transformer) and six datasets (MNIST, CIFAR-10, CIFAR-100, Fashion-MNIST, IMDB, Moving MNIST). NDNA consistently outperforms random sparsity at matched density by 0.39% to 21.7% and matches or exceeds dense baselines on five of six tasks. On Moving MNIST video prediction, a factored spatiotemporal genome of 374 parameters matches the dense ceiling (MSE 62.23 vs 62.15) while random sparsity at the same 11.1% density scores 79.44, a 21.7% relative gap. On IMDB, the genome transformer (85.05%) beats the standard dense transformer (84.57%) while using only 22.1% of possible connections. Topology transfers across tasks: a CIFAR-10 trained genome achieves 60.92% on CIFAR-100 without retraining, compared to 53.91% for random sparse at matched density. Across all experiments, 226 to 374 parameters control up to 2.2 million connections (8,384:1 compression). The existence of a compact, transferable representation of useful network topology suggests that optimal neural architectures occupy a low-dimensional manifold navigable by simple developmental programs.

1. Introduction

Neural network performance depends critically on architecture. Residual connections (He et al., 2016), multi-head attention (Vaswani et al., 2017), and efficient scaling rules (Tan and Le, 2019) each delivered step-function improvements by changing how information flows through networks. Yet architecture design remains either manual, requiring deep expertise, or automated via neural architecture search (NAS), requiring thousands of GPU hours per task (Zoph and Le, 2017).

Biology solved this problem differently. The human genome encodes roughly 20,000 genes in 3 billion base pairs, yet specifies a brain with 86 billion neurons and 100 trillion synapses. DNA does not store synaptic weights or enumerate individual connections. It encodes developmental programs: rules for cell differentiation, axon guidance, and synaptic formation that grow neural circuits from a compact specification. The genome is a compression of architecture, not of weights.

We take this observation literally. We introduce Neural DNA (NDNA), a learned genome of fewer than 300 parameters that specifies how neural network connections form. The genome encodes cell type embeddings and a compatibility matrix. During training, these produce binary connectivity masks that gate every information path in the network. Connections that the genome does not grow do not exist. A metabolic cost term penalizes total connectivity, pressuring the genome toward sparse, efficient topologies.

Three design principles distinguish NDNA from prior work on sparse networks and indirect encodings:

1. **Default disconnected.** All connections start inactive. The genome must actively grow each connection, analogous to axonal growth in biological development. This contrasts with pruning approaches (LeCun et al., 1989; Han et al., 2015) that start fully connected and remove weights.

2. **Type-based compatibility.** Neurons are assigned continuous type distributions based on their band (layer group) and position. Connection probability depends on the compatibility between source and target types, enabling the genome to express structured wiring patterns (e.g., "all early-layer neurons of type A connect to all late-layer neurons of type B") without enumerating individual connections.

3. **Metabolic cost.** A sparsity loss penalizes the total strength of grown connections, analogous to the metabolic cost of maintaining biological synapses. This prevents the genome from trivially growing all connections and forces it to allocate connectivity where it matters most.

We evaluate NDNA across four architectures and six datasets, including video prediction, the most compute-intensive domain in modern AI. Video generation models process millions of spatiotemporal tokens with quadratic attention cost; structured sparsity that preserves performance while eliminating unnecessary connections could dramatically reduce that cost. The genome consistently discovers useful topology: it beats random sparsity at matched density on every experiment, matches or exceeds dense baselines on most, and transfers topology across tasks. A genome trained on CIFAR-10 grows networks

for CIFAR-100 that outperform random wiring by 7.01 percentage points. On Moving MNIST video prediction, a factored spatiotemporal genome of 374 parameters matches the dense baseline within 0.1% while random sparsity at the same density is 21.7% worse, the largest gap in any experiment.

The core contribution is empirical evidence that useful network topology can be compressed into a developmental program of fewer than 400 parameters, that this program generalizes across tasks and modalities (static images, text, video), and that the resulting topology outperforms both random wiring and dense connectivity on multiple benchmarks.

2. Related Work

Neural Architecture Search

Neural architecture search automates the discovery of network architectures. Zoph and Le (2017) used reinforcement learning to search over architectures, requiring 800 GPUs for 28 days. DARTS (Liu et al., 2019) introduced differentiable search, reducing cost by orders of magnitude but still requiring per-task optimization. EfficientNet (Tan and Le, 2019) combined NAS with compound scaling rules to achieve state-of-the-art vision results. All NAS methods search for a fixed architecture per task. NDNA learns a developmental program that generates topology, enabling transfer across tasks without re-search.

Sparse Networks and Pruning

The Lottery Ticket Hypothesis (Frankle and Carbin, 2018) demonstrated that dense networks contain sparse subnetworks that, when trained in isolation from their original initialization, match full network accuracy. Optimal Brain Damage (LeCun et al., 1989) and magnitude pruning (Han et al., 2015) remove weights from trained networks based on importance scores. RigL (Evci et al., 2020) and SET (Mocanu et al., 2018) train sparse networks directly by periodically reallocating connections. Hoefler et al. (2021) provide a comprehensive survey of sparsity methods, establishing dynamic sparse training (DST) as a general paradigm encompassing both pruning and growth strategies. These methods operate on individual weight-level sparsity. NDNA generates structured sparsity patterns from a compact program, enabling topology transfer and compression ratios orders of magnitude higher than weight-level masks.

Indirect Encoding and Neuroevolution

NEAT (Stanley and Miikkulainen, 2002) evolves network topology and weights simultaneously through complexification. HyperNEAT (Stanley et al., 2009) uses compositional pattern-producing networks (CPPNs) to generate weight values as a function of spatial position, enabling large-scale network generation from compact specifications. Weight Agnostic Neural Networks (Gaier and Ha, 2019) search for topologies that perform well regardless of weight values. NDNA differs from HyperNEAT in generating topology masks rather than weight values, a distinction we found critical (Section 5.2 and Appendix A). Unlike WANN, NDNA uses gradient-based learning rather than evolutionary search, and controls topology through cell type compatibility rather than direct edge specification.

Developmental Neural Networks

Cellular encoding (Gruau, 1994) and developmental programs (Miller, 2004) explored biologically-inspired network growth through grammar-based or cellular automaton rules, using evolutionary optimization. Najarro and Risi (2020) meta-learned Hebbian plasticity rules that generate network connectivity. These approaches relied on evolutionary search over the developmental program. NDNA uses gradient descent with a differentiable genome, enabled by the straight-through estimator for binary mask generation, and jointly optimizes the developmental program with network weights.

3. Method

3.1 Overview

NDNA consists of a compact genome G with parameters θ_G that generates binary connectivity masks M for a target network N with weights θ_N. The genome encodes type embeddings and compatibility rules. Neurons are organized into bands (analogous to layer groups). The mask for each band pair is computed from the type distributions of source and target neurons and their compatibility, producing a structured sparsity pattern that the genome can learn to optimize.

3.2 Genome Architecture

The genome contains six parameter groups:

1. **Type affinity matrix** $A \in \mathbb{R}^{K \times D}$: continuous embeddings for K cell types in D dimensions.
2. **Compatibility matrix** $C \in \mathbb{R}^{K \times K}$: direct type-to-type connection preference.
3. **Connection scale** $\gamma \in \mathbb{R}$: learned global scaling factor.
4. **Depth penalty** $\delta \in \mathbb{R}$: learned cost for cross-band connections.
5. **Band type base** $B_{\text{base}} \in \mathbb{R}^{L \times K}$: base type logits per band.
6. **Band type gradient** $B_{\text{grad}} \in \mathbb{R}^{L \times K}$: positional variation of type logits within each band.

With $K = 8$, $D = 8$, and L bands, the total parameter count is $KD + K^2 + 1 + 1 + 2LK$. For $L = 6$ (MLP): 226 parameters. For $L = 8$ (CNN, Transformer): 258 parameters.

Type distribution. Each neuron j at position $p_j \in [0, 1]$ within band b has a type distribution:

$$\tau_j = \text{softmax}\left(3 \cdot \left(B_{\text{base}}[b] + p_j \cdot B_{\text{grad}}[b]\right)\right) \in \mathbb{R}^K \tag{1}$$

where $p_j = j/(n_b - 1)$ with n_b neurons in band b. The factor of 3 sharpens type assignments toward categorical distributions.

Connection rule. The type-to-type connection logit matrix is:

$$R = \left(\frac{AA^\top}{\sqrt{D}} + C\right) \cdot \text{softplus}(\gamma) \in \mathbb{R}^{K \times K} \tag{2}$$

The affinity term $AA^\top/\sqrt{D}$ captures similarity between type embeddings. The compatibility matrix C adds a learned bias. The scaling factor $\text{softplus}(\gamma)$ controls overall connection strength.

Mask generation. The connectivity mask between source band s and target band t with n_s and n_t neurons respectively is:

$$\Lambda_{st} = T_t \cdot R \cdot T_s^\top \in \mathbb{R}^{n_t \times n_s} \tag{3}$$

where $T_s \in \mathbb{R}^{n_s \times K}$ and $T_t \in \mathbb{R}^{n_t \times K}$ are the type distribution matrices from Equation 1. A depth penalty is subtracted:

$$\Lambda_{st} \leftarrow \Lambda_{st} - \text{softplus}(\delta) \cdot \frac{|t-s|}{L} \tag{4}$$

The final soft mask is:

$$M_{st} = \sigma(\Lambda_{st} \cdot \alpha) \in [0,1]^{n_t \times n_s} \tag{5}$$

where α is a temperature parameter annealed from 1.0 to 10.0 during training. At $\alpha = 10$, the sigmoid produces near-binary outputs (>0.99 or <0.01).

Hard masks. For binary {0, 1} masks during forward pass, we use a straight-through estimator (Bengio et al., 2013):

$$M_{st}^{\text{hard}} = \mathbb{1}[\sigma(\Lambda_{st}) > 0.5] \tag{6}$$

with gradients computed as $\nabla_\Lambda M^{\text{hard}} = \sigma(\Lambda)(1 - \sigma(\Lambda))$.

3.3 Default Disconnected Initialization

The compatibility matrix C is initialized as $C_{ij} \sim \mathcal{N}(-1.0, 0.3^2)$. This negative initialization means that all type pairs are initially incompatible, producing negative logits Λ_{st} and near-zero masks M_{st}. The network starts disconnected. The genome must learn to increase specific compatibility entries to grow connections.

This design choice is critical. In early experiments (Appendix A), we initialized C near zero (default connected or neutral). The genome consistently converged to the trivial solution of growing all connections, producing topology indistinguishable from a dense network. Default-disconnected initialization forces the genome to justify every connection through improved task performance, producing genuinely sparse and structured topologies.

3.4 Metabolic Cost (Sparsity Loss)

A metabolic cost term penalizes the total strength of grown connections:

$$\mathcal{L}_{\text{sparse}} = \frac{\sum_{t>s} \sum_{i,j} M_{st}[i,j]}{\sum_{t>s} n_t \cdot n_s} \tag{7}$$

This is the mean mask value across all possible connections. It is differentiable through the soft mask M_{st} and provides gradient signal to the genome to reduce connectivity.

3.5 Depth Penalty

The depth penalty in Equation 4 makes long-range connections (between distant bands) more costly than short-range connections. The penalty magnitude $\text{softplus}(\delta)$ is learned, allowing the genome to control how aggressively it discourages long-range wiring. This mirrors the biological observation that axonal growth has metabolic and signaling costs proportional to distance.

3.6 Training Procedure

The genome parameters θ_G and network weights θ_N are optimized jointly with separate optimizers:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \cdot \mathcal{L}_{\text{sparse}} \tag{8}$$

where $\mathcal{L}_{\text{task}}$ is cross-entropy and λ is the sparsity weight. Network weights use task-appropriate optimizers (Adam for MLPs, SGD with momentum for CNNs, AdamW for transformers). Genome parameters use Adam with a higher learning rate (0.001 to 0.01) to enable rapid topology exploration. Temperature α is annealed linearly from 1.0 to 10.0 over training, smoothly transitioning from soft to hard masks. Full hyperparameters are in Appendix B.

3.7 Architecture-Specific Instantiation

MLP. Bands map to neuron groups: input, hidden layers, output. Masks M_{st} are applied element-wise to weight matrices W_{st}. For a forward pass through band t: $h_t = \text{ReLU}(\sum_{s<t}(M_{st} \odot W_{st})^\top x_s + b_t)$. All band pairs (s, t) with $s < t$ have masks, enabling skip connections to emerge.

CNN. Bands map to convolutional layer groups (input, six conv layers in three resolution groups, classifier). Masks $M_{st} \in \{0, 1\}^{C_{\text{out}} \times C_{\text{in}}}$ gate channel-to-channel connectivity. For 3x3 convolutions, the mask is broadcast: $W_{\text{masked}} = W \odot M[:, :, \text{None}, \text{None}]$, zeroing entire 3x3 kernels for masked channel pairs. Skip connections between non-adjacent bands use 1x1 convolutions with adaptive average pooling for spatial alignment. Hard binary masks (straight-through estimator) are used throughout.

Transformer. Bands map to the embedding layer, six transformer layers, and the classifier. We replace PyTorch's `nn.MultiheadAttention` with manual multi-head attention to enable genome masking of the output projection. Query, key, and value projections and the attention computation itself are standard (no genome involvement). The genome masks three components per layer:

1. **Attention output projection W_O**: mask shape (H, H) where H is hidden dimension. This controls how head outputs combine back to the hidden representation.
2. **Feed-forward first linear**: mask shape (F, H) where F is the FF intermediate dimension. This controls which FF neurons activate.
3. **Cross-layer skip projections**: mask shape (H, H) for non-adjacent band pairs.

The classifier layer from the final transformer layer to output logits is also genome-masked. Every information path passes through a genome gate. There are no unmasked (free) paths.

Video Transformer. Input frames (64x64 grayscale) are divided into an 8x8 grid of 8x8 pixel patches, yielding 64 patches per frame. Each patch is linearly projected to the hidden dimension (128) and summed with learned spatial and temporal positional embeddings. The architecture uses divided space-time attention (Bertasius et al., 2021), where each layer performs spatial attention (64 patches within each frame), then temporal attention (10 frames at each spatial position), then a feed-forward block. The genome controls three types of attention masking:

1. **Temporal attention mask**: shape (T, T) where $T = 10$ frames. Controls which frame pairs attend to each other at every spatial position.
2. **Spatial attention mask**: shape (P, P) where $P = 64$ patches. Controls which patch pairs attend to each other within every frame.
3. **Depth mask**: shape (H, H) for the output projection W_O within each layer, and cross-layer skip projections. Same mechanism as the IMDB transformer.

The temporal and spatial masks are applied as pre-softmax additive biases: the mask value $m \in [0.01, 0.99]$ is mapped to a bias $(m - 0.5) \cdot \beta$ where $\beta = 20$, producing biases in $[-10, +10]$ that gate attention scores before softmax. This ensures position-specific gradient flow: each mask entry affects only its corresponding attention pair.

3.8 Factored Spatiotemporal Extension

For video, a single genome's type system would need to encode both "which frame" and "which patch" in one type vector, conflating temporal and spatial structure. We factor the genome into three independent sub-genomes:

1. **Temporal genome** G_t: $K_t = 4$ types, $D_t = 4$ dimensions, $L_t = 5$ bands. 74 parameters. Generates the (T, T) temporal attention mask via $M_t = G_t.\mathrm{growth_mask}(0, 1, T, T)$.
2. **Spatial genome** G_s: $K_s = 4$ types, $D_s = 4$ dimensions, $L_s = 5$ bands. 74 parameters. Generates the (P, P) spatial attention mask via $M_s = G_s.\mathrm{growth_mask}(0, 1, P, P)$.
3. **Depth genome** G_d: $K_d = 8$ types, $D_d = 8$ dimensions, $L_d = 6$ bands. 226 parameters. Generates the depth/cross-layer masks as in the standard transformer. The depth genome uses the default initialization (compatibility $\sim \mathcal{N}(-1.0, 0.3^2)$).

Total genome parameters: $74 + 74 + 226 = 374$.

Each sub-genome uses the same growth mask mechanism from Equations 1-6, but with independent parameters. The factored structure has two advantages. First, it separates temporal patterns (which frames matter) from spatial patterns (which patches matter), allowing each to be learned without interference. Second, it is more parameter-efficient: a single genome with $K = 8$ types covering all $T \times P = 640$ tokens would produce a 640×640 mask, requiring the type system to encode both temporal position and spatial position in one vector. Factoring lets two smaller genomes of 4 types each learn 10x10 and 64x64 masks independently.

The temporal and spatial sub-genomes use modified initialization to prevent sigmoid saturation: compatibility C initialized to 0 (neutral, rather than -1 as in the depth genome), connection scale $\gamma = 1.0$ (rather than 3.0), and band type parameters scaled to 20% of their default magnitude. Mask outputs are clamped to $[0.01, 0.99]$ to maintain gradient flow. Metabolic cost is applied only to the depth genome; the temporal and spatial genomes learn freely without sparsity pressure, since their masks are small enough (100 and 4,096 entries respectively) that sparsity is not needed for efficiency.

4. Experiments

4.1 Experimental Setup

All experiments run on an Apple M3 MacBook Pro (8GB) using the MPS backend. Random seed 42 is fixed across all experiments. We compare four models in each experiment:

1. **Dense baseline**: standard architecture (MLP, ResNet, Transformer, or Video Transformer) with no sparsity.
2. **NDNA (Genome)**: the proposed method.
3. **Random Sparse**: same architecture as NDNA but with fixed random binary masks at matched density. The control that isolates the effect of learned vs random topology.
4. **Dense Skip**: same skip-connection architecture as NDNA but with all masks set to 1.0 (no sparsity). Shows what the full skip architecture achieves without the genome's selective wiring.

4.2 Transformer Experiments

IMDB Sentiment Classification. Six-layer transformer encoder, hidden dimension 256, 4 attention heads, FF dimension 512, sequence length 512. Eight bands (embedding, 6 layers, classifier). 258 genome parameters. Split optimizer: AdamW (lr=2e-4) for weights, Adam (lr=0.01) for genome. 15 epochs, $\lambda = 0.005$. Temperature annealed 1.0 to 10.0.

Table 1: Transformer Results on IMDB

Model	Params	Accuracy
Dense Transformer	11,108,354	84.57%
Dense Skip Transformer	12,090,882	84.68%
NDNA Transformer	**12,089,604**	**85.05%**
Random Sparse (22.1%)	12,089,346	84.66%

NDNA beats the dense transformer ceiling by **+0.48%** and random sparse by **+0.39%**. The genome discovered a non-trivial topology:

Table 2: Transformer Layer Connectivity (NDNA on IMDB)

Connection	Type	Hard Density	Soft Density
emb to L1	Attention W_O	100.0%	99.9%
emb to L1	FF	100.0%	99.9%
L1 to L2	Attention W_O	0.0%	37.0%
L1 to L2	FF	0.0%	37.0%
emb to L3	Skip	100.0%	96.8%
L3 to L4	Attention W_O	0.0%	19.9%
L3 to L4	FF	0.0%	19.9%
L4 to L5	Attention W_O	51.4%	48.4%
L4 to L5	FF	51.4%	48.4%
L6 to cls	Classifier	100.0%	100.0%

The genome built a highway: full connectivity at entry (emb to L1), a 100% skip from embedding directly to L3 (bypassing L2 entirely), selective half-connectivity at L4 to L5, and full connectivity to the classifier. Layers 2, 5, and 6 received minimal or no new connections. The genome effectively reduced a 6-layer transformer to approximately 3 active layers with a skip highway, a structural simplification that improved accuracy.

4.3 Transfer Experiments

CIFAR-10 to CIFAR-100. We freeze the CIFAR-10 trained genome and use its topology to train a network on CIFAR-100 (100 classes, 500 samples per class). Only network weights are trained; topology is fixed. 258 genome parameters. 250 epochs, SGD with momentum, cosine annealing.

Table 3: CNN Transfer from CIFAR-10 to CIFAR-100

Model	Params	Accuracy	Notes
Dense ResNet	181,108	67.16%	Ceiling
Fresh Genome (CIFAR-100)	92,966	61.10%	Genome + weights trained on C100
Frozen Genome (CIFAR-10)	92,966	60.92%	Genome frozen from C10
Random Sparse (47.6%)	92,708	53.91%	Matched density

The frozen CIFAR-10 genome achieves **60.92%** on CIFAR-100, within 0.18% of a fresh genome trained from scratch on CIFAR-100 (61.10%). Both genome variants beat random sparse by **+7.01%** and **+7.19%** respectively. The topology learned on 10 classes transfers to 100 classes with negligible loss, while random wiring at the same density loses 7 percentage points.

Topology convergence. The CIFAR-10 genome (41.0% hard density) and independently trained CIFAR-100 genome (40.2% hard density) converge to similar overall density and share structural features: dense back-end connectivity, sparse front-end, and selective skip connections. The CIFAR-100 genome additionally grows a long-range skip from input to group 3, likely needed for the finer-grained 100-class discrimination.

Fashion-MNIST to MNIST. In a separate MLP transfer experiment, a genome trained on Fashion-MNIST (clothing images) was frozen and used to grow a network for MNIST (digit images). The frozen genome achieved 97.54% on MNIST, compared to 97.37% for a fresh MNIST genome and 11.35% for random sparse at matched density. The genome learned general structural principles (e.g., where to place skip connections, which layers to keep sparse) that transfer across visual domains.

4.4 CNN Experiments

CIFAR-10. Eight bands: input (3ch), two conv groups of 16ch, two of 32ch, two of 64ch, and a 10-class FC layer. 258 genome parameters. Hard binary masks via straight-through estimator. Split optimizer: SGD with momentum (lr=0.1) for CNN weights, Adam (lr=0.001) for genome. 200 epochs, $\lambda = 0.01$. Temperature annealed from 1.0 to 10.0.

Table 4: CNN Results on CIFAR-10

Model	Params	Accuracy	Density
Dense ResNet	175,258	89.80%	100%
Dense Skip CNN	86,858	88.71%	100%
NDNA CNN	**87,116**	**88.93%**	**44.6%**
Random Sparse CNN (44.6%)	86,858	85.78%	44.6%

NDNA beats random sparse by **+3.15%** at matched 44.6% density. Despite having half the parameters of Dense ResNet, the genome CNN comes within 0.87% of the ceiling. The genome discovered structured connectivity: dense back-end layers (groups 2 and 3 at near-100% density), sparse front-end (group 1 at lower density), and selective medium-range skip connections. This pattern mirrors hand-designed architectures where later layers need richer feature combinations.

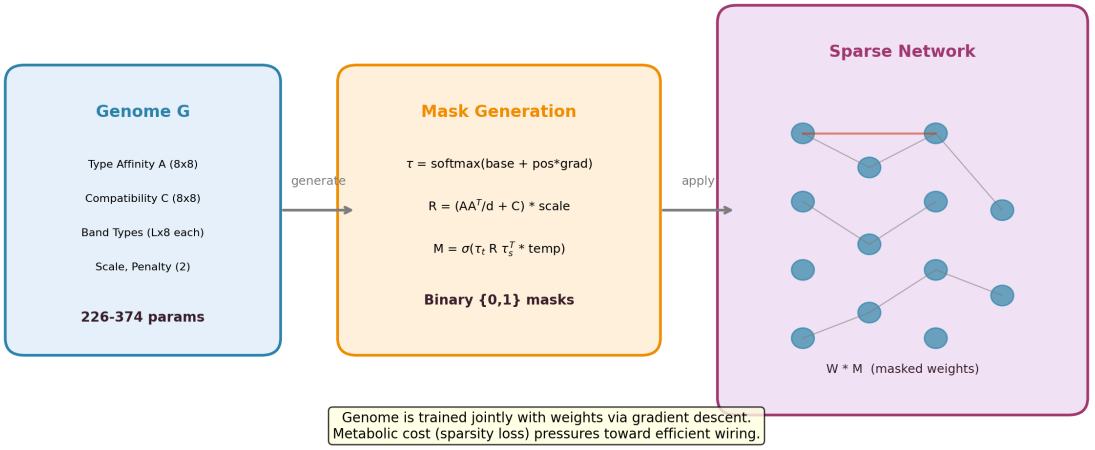

Figure 1: NDNA method overview. A compact genome (226-258 parameters) generates binary connectivity masks through type-based compatibility rules, which are applied to network weights. The genome is trained jointly with weights via gradient descent, with metabolic cost pressure toward efficient wiring.

4.5 MLP Experiments

MNIST. Input dimension 784 (28x28 grayscale). Four hidden bands of 48 neurons each (6 bands total, 226 genome parameters). 25 epochs, Adam optimizer, cosine annealing, sparsity weight $\lambda = 0.1$.

Table 5: MLP Results on MNIST

Model	Params	Accuracy	Density
Normal MLP (h=128, 2 layers)	118,282	98.33%	100%
Dense Skip	174,314	98.05%	100%
NDNA (Genome)	**174,540**	**97.54%**	**11.6%**
Random Sparse (11.6%)	174,314	97.09%	11.6%

NDNA beats random sparse by **+0.45%** at 11.6% soft density. The genome (226 params) controls 174,112 possible connections at 770:1 compression. The genome achieves 97.54% using only 3,816 active connections (2.2% hard density), within 0.79% of the dense MLP ceiling.

CIFAR-10 (MLP). Input dimension 3072 (32x32x3 color). Four hidden bands of 128 neurons (6 bands, 226 genome parameters). 40 epochs, Adam, cosine annealing, $\lambda = 0.1$.

Table 6: MLP Results on CIFAR-10

Model	Params	Accuracy	Density
Normal MLP (h=256, 3 layers)	920,842	54.32%	100%
Dense Skip	1,707,530	51.27%	100%
NDNA (Genome)	**1,707,756**	**57.14%**	**4.6%**
Random Sparse (4.6%)	1,707,530	51.68%	4.6%

On the harder CIFAR-10 task, the genome advantage grows. NDNA beats random sparse by **+5.46%** and beats the dense MLP by **+2.82%**. Notably, the dense skip network (all connections active) performs worst, suggesting that unstructured full connectivity hurts on this task. The genome's selective wiring finds paths that neither random masks nor dense connectivity discover. Compression ratio: 7,553:1 (226 genome params controlling 1,707,008 connections).

4.6 Video Experiments

Moving MNIST Next-Frame Prediction. Four-layer divided space-time transformer, hidden dimension 128, 4 attention heads, FF dimension 256. Input: 10 frames of 64x64 grayscale with 2 bouncing MNIST digits. Output: predicted frame 11. 5,000 training sequences, 1,000 test sequences, procedurally generated. 374 genome parameters (74 temporal + 74 spatial + 226 depth). Split optimizer: AdamW (lr=3e-4) for weights, Adam (lr=0.01) for depth genome, Adam (lr=0.01) for temporal/spatial genomes. 50 epochs for genome, 30 for baselines. $\lambda = 0.005$. Temperature annealed 1.0 to 10.0. Loss: MSE (lower is better).

Table 8: Video Transformer Results on Moving MNIST

Model	Params	MSE
Dense Video Transformer	820,416	62.154
Dense Skip Video Transformer	918,720	62.090
NDNA Video Transformer	**919,094**	**62.226** (genome: 374)
Random Sparse (11.1%)	918,720	79.443

NDNA beats random sparse by **21.7% relative MSE** (62.23 vs 79.44) at matched 11.1% density, the largest genome-vs-random gap in any experiment. The genome matches the dense ceiling within 0.1% (62.23 vs 62.15). The factored genome discovered structured connectivity in both temporal and spatial dimensions.

Temporal connectivity. The genome learned a clear recency pattern in temporal attention. Recent source frames (8, 9) receive attention weights of 0.90-0.91, while distant frames (0-5) receive weights below 0.01. The genome independently discovered that for next-frame prediction, the most recent

frames carry the most predictive information, with attention decaying smoothly toward earlier frames. This mirrors the temporal redundancy principle used by video codecs (H.264, H.265) for inter-frame compression, but was learned automatically by the genome from 374 parameters.

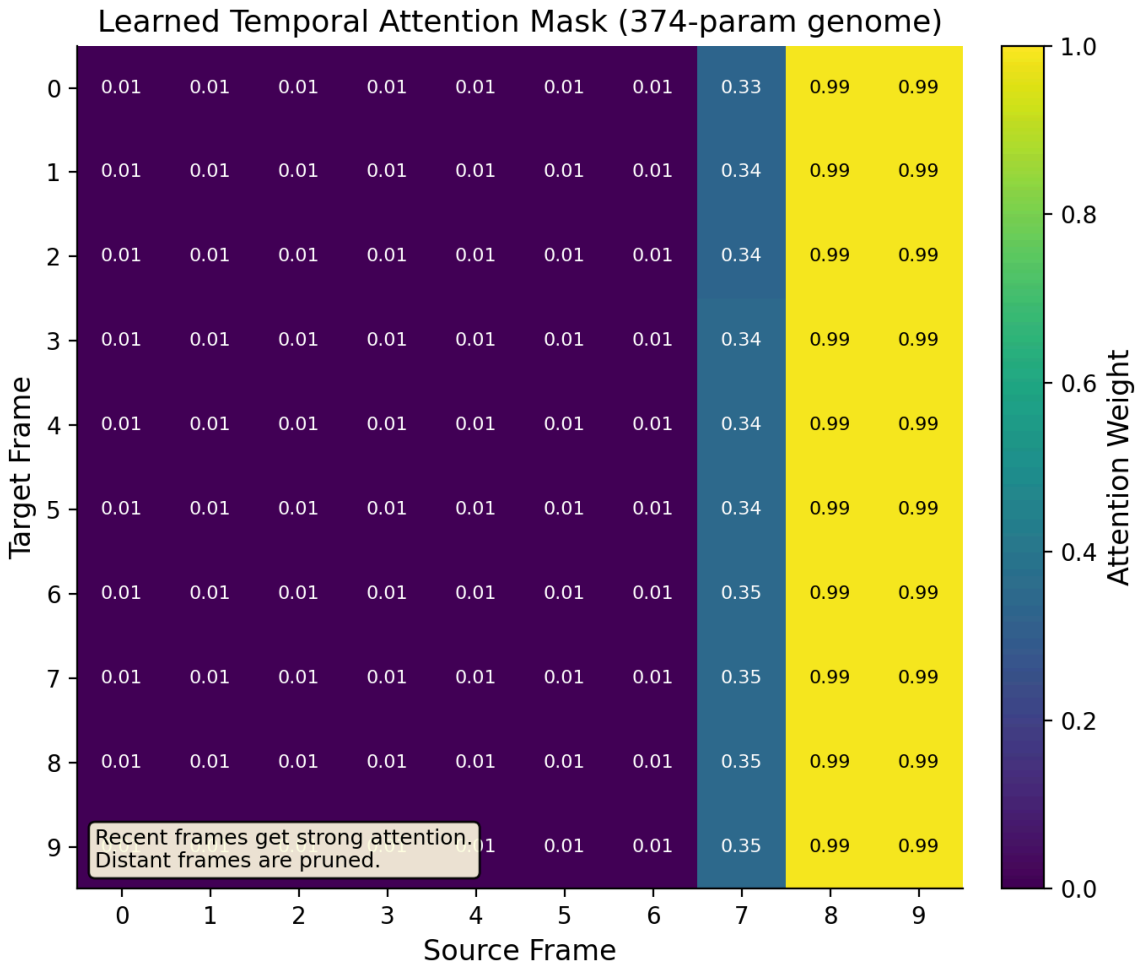

Figure 7: Learned temporal attention mask on Moving MNIST. Each column represents a source frame. The genome learned that recent frames (8-9) should receive strong attention at all target positions, while distant frames (0-5) are effectively pruned. This recency pattern was discovered automatically.

Spatial connectivity. The spatial mask exhibits monotonic decay with Manhattan distance between patches. Nearby patches (distance 0-2) receive attention strength of 0.47-0.48, while the most distant patches (distance 14) receive 0.30. The genome discovered spatial locality without any architectural bias toward it.

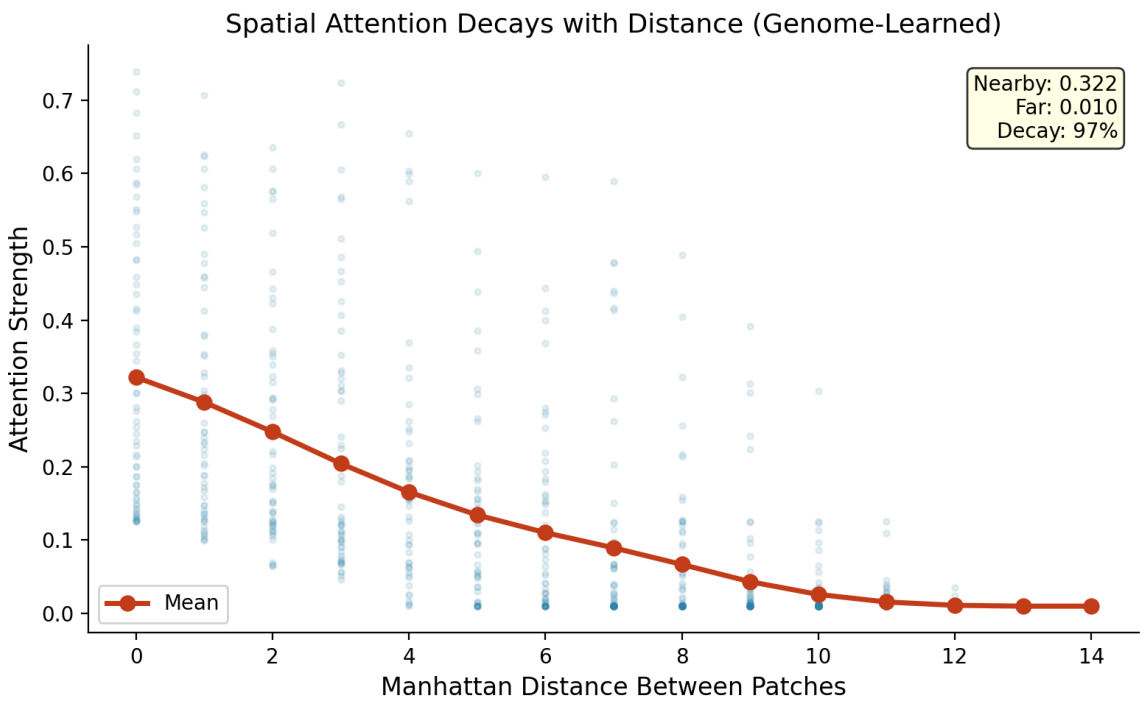

Figure 8: Spatial attention strength as a function of Manhattan distance between patches. The genome learns spatial locality: nearby patches connect more strongly than distant patches. Each blue dot is one patch pair; the red line shows the mean.

Gap grows with complexity. The genome-vs-random gap increases monotonically with task complexity: 0.45% on MNIST, 3.15% on CIFAR-10 CNN, 5.46% on CIFAR-10 MLP, 7.01% on CIFAR-100 transfer, and 21.7% on video. As tasks become harder and the space of possible topologies grows, the advantage of learned over random connectivity compounds.

4.7 Compression Analysis

Table 9: Compression Ratios Across All Experiments

Experiment	Genome Params	Possible Connections	Compression
MNIST MLP	226	174,112	770:1
CIFAR-10 MLP	226	1,707,008	7,553:1
CIFAR-10 CNN	258	22,654	88:1
CIFAR-100 CNN	258	43,084	167:1
IMDB Transformer	258	2,163,200	8,384:1
Moving MNIST Video	374	307,300	821:1

Compression ratio scales with network size. The genome ranges from 226 to 374 parameters depending on architecture. The video experiment uses 374 parameters because the factored spatiotemporal extension adds separate temporal (74) and spatial (74) sub-genomes alongside the standard depth genome

(226). As networks grow (from 174K to 2.2M connections), the compression ratio increases from 770:1 to 8,384:1. This scaling property means NDNA becomes more efficient on larger networks, the opposite of weight-level pruning where mask overhead grows linearly with network size.

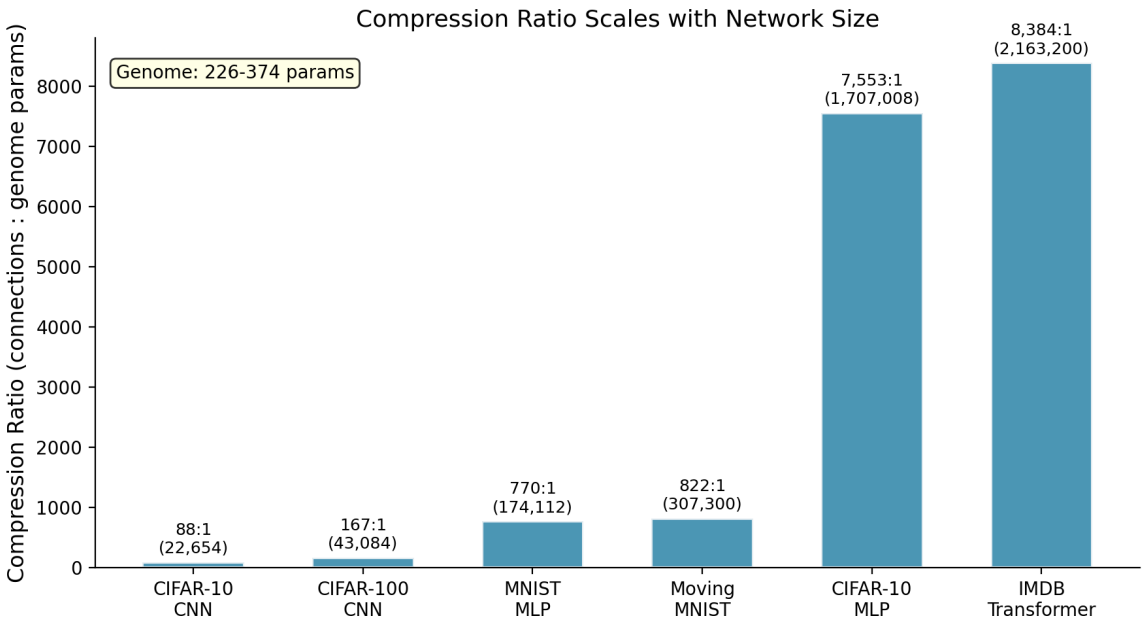

Figure 2: Compression ratio (possible connections per genome parameter) scales with network size. The genome ranges from 226 to 374 parameters. Larger networks achieve higher compression.

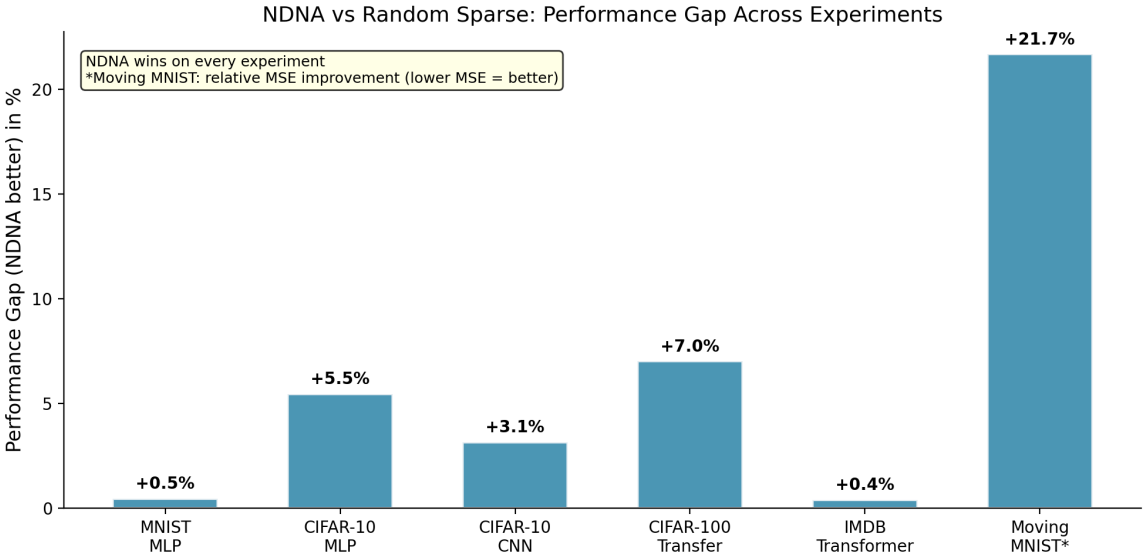

Figure 3: Performance gap (NDNA minus Random Sparse) across all experiments. NDNA outperforms random sparsity on every task, with larger gaps on harder problems. The video experiment (21.7% relative MSE improvement) produces the largest gap.

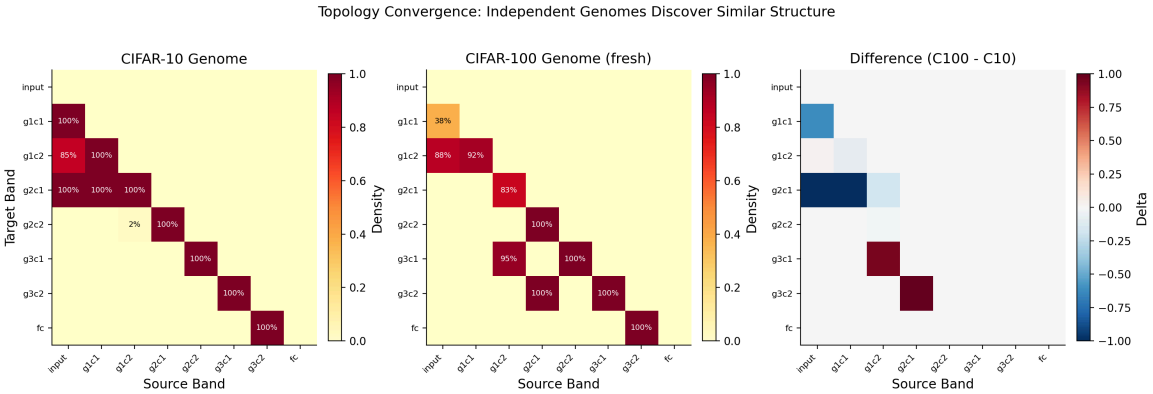

Figure 4: Topology convergence. CIFAR-10 and CIFAR-100 genomes, trained independently, discover similar connectivity structure. The difference matrix (right) shows the genomes agree on most band pairs.

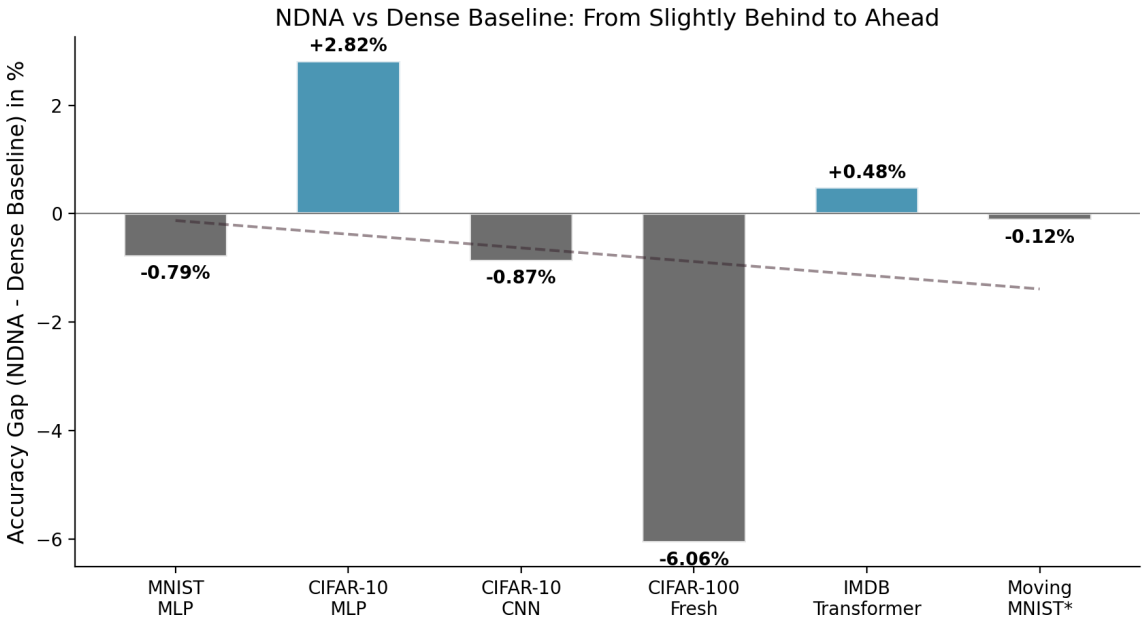

Figure 5: NDNA accuracy relative to the dense baseline across experiments. The genome progresses from slightly below dense on easy tasks (MNIST) to above dense on harder tasks (CIFAR-10 MLP, IMDB). Note: the CIFAR-100 comparison (-6.06%) is between the genome CNN (93K params) and the Dense ResNet (181K params), a 2x parameter mismatch. The gap reflects the parameter budget difference, not a topology failure.

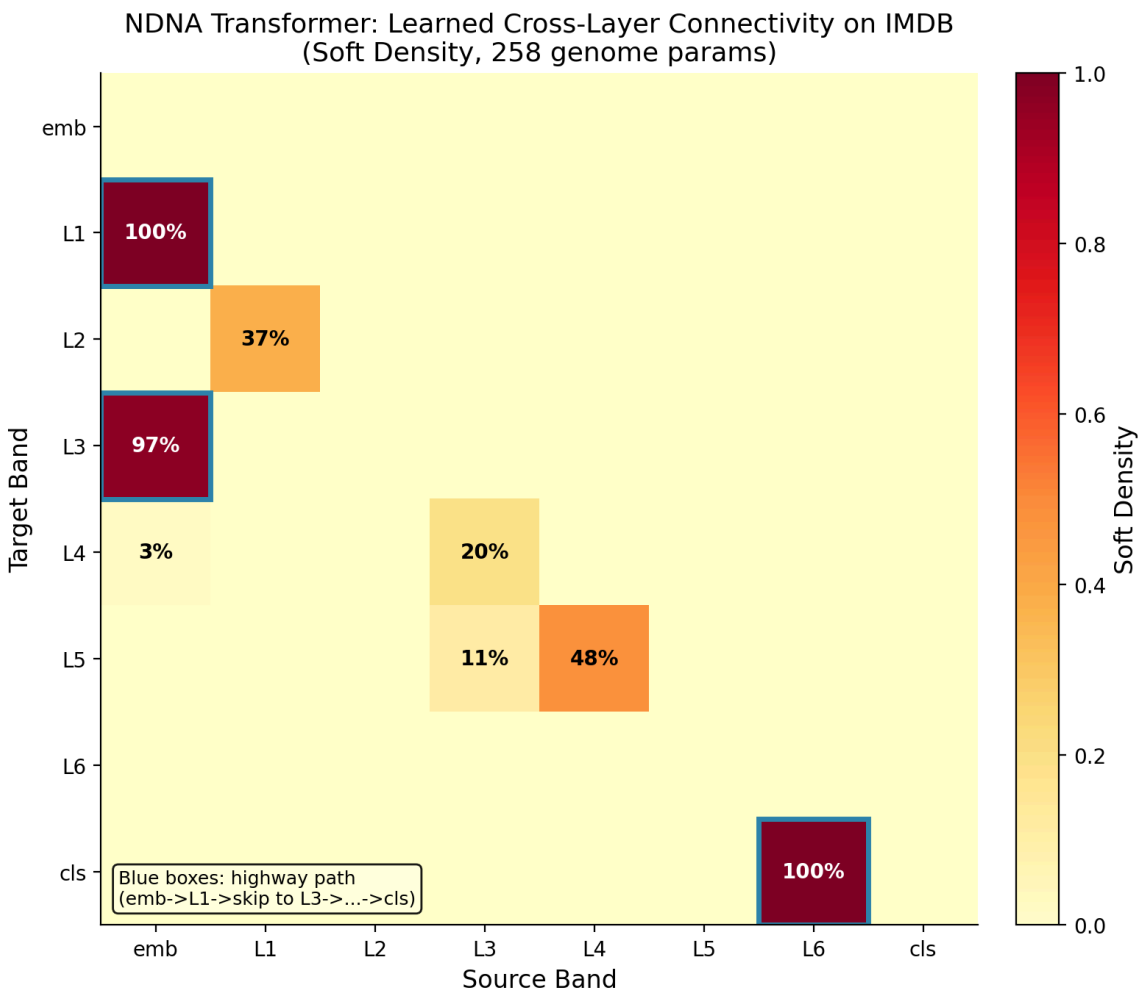

Figure 6: Learned transformer cross-layer connectivity on IMDB. Dark cells indicate active connections. The genome builds a highway from embedding through L1 and L3, bypasses L2 and most of L5/L6, and maintains full connectivity to the classifier.

5. Analysis

5.1 What the Genome Learns

Several structural principles emerge consistently across architectures:

1. **Entry and exit layers are fully connected.** The genome always maintains 100% connectivity at the input and output boundaries, ensuring all features enter and all classes are accessible.

2. **Interior layers are selectively sparse.** Middle layers receive partial or zero connectivity. The genome discovers that not all layers contribute equally and prunes information paths through less useful ones.

3. **Skip connections emerge where useful.** The CNN genome grows medium-range skip connections (e.g., group 1 to group 2, bypassing intermediate layers). The transformer genome grows a direct skip from the embedding to layer 3, bypassing layer 2 entirely.

4. **Density converges to 15% to 45%.** Across all experiments, the genome settles on densities well below 50%, indicating that networks are significantly overparameterized in their default connectivity.

5. **Attention and feed-forward masks correlate.** In the transformer, attention W_O and FF masks for the same layer pair have identical densities, suggesting the genome treats each layer as a unit rather than independently controlling subcomponents.

6. **Temporal connectivity follows recency.** In the video experiment, the genome learned that recent frames (8-9) receive attention weights of 0.90-0.91 while distant frames (0-5) receive weights below 0.01. This recency decay was discovered automatically from 74 temporal genome parameters, mirroring the temporal redundancy principle used by video codecs.

7. **Spatial locality emerges.** The spatial genome learned to connect nearby patches more densely (attention strength 0.48 at distance 0) than distant patches (0.30 at distance 14), discovering spatial locality without any architectural bias toward it. No convolutional structure, kernel sizes, or locality constraints were imposed.

5.2 Why Default Disconnected Matters

Default-disconnected initialization was the critical design choice that made NDNA work. In seven preceding experimental phases spanning approximately 60 hours of compute and iteration (Appendix A), every approach that started with default-connected or neutral initialization failed to produce meaningful topology. The genome would either converge to full connectivity (when compatible) or remain near its initialization (when neutral).

The asymmetry is in gradient dynamics. From a connected state, the gradient signal to "remove this connection" is weak: removing one connection among thousands has negligible effect on loss. From a disconnected state, the gradient signal to "grow this connection" is strong: adding connectivity to a bottleneck dramatically reduces loss. Default disconnected creates a strong selection pressure for useful connections and no pressure for useless ones.

5.3 Transfer as Evidence of Universal Topology

The CIFAR-10 to CIFAR-100 transfer result is the strongest evidence that the genome learns general structural principles rather than task-specific wiring. Key observations:

1. The frozen genome (60.92%) nearly matches the fresh genome (61.10%), a gap of only 0.18%.
2. Both independently-trained genomes converge to similar density (41.0% vs 40.2%) and similar structural patterns.
3. Random sparse at matched density scores 53.91%, a gap of 7.01% from the frozen genome.

If the genome were memorizing CIFAR-10-specific connectivity, it should fail on CIFAR-100's 100-class fine-grained discrimination task. Instead, the topology transfers because it encodes domain-general principles: "early layers should be sparse, late layers should be dense, certain skip connections help." These principles apply across visual classification tasks.

5.4 Limitations

We identify six limitations of the current work:

1. **Small-scale benchmarks.** All experiments use networks with fewer than 12 million parameters on small datasets. The video experiment uses Moving MNIST (64x64, bouncing digits), not natural video. Whether NDNA scales to billion-parameter models on real video (1080p, natural scenes, longer sequences) is unproven. The compression ratio trend (increasing with network size) is encouraging but not demonstrated at scale.

2. **Limited architectural control.** The genome controls topology (which connections exist) but not width (how many neurons per layer), activation functions, or normalization choices. A more complete developmental program would control these additional axes.

3. **Modest margins on some tasks.** The IMDB transformer gap of 0.39% over random sparse, while consistent with gaps on other tasks, is small enough that statistical significance would require multi-seed experiments.

4. **Single seed.** All experiments use a single random seed (42). Multi-seed runs with variance reporting are planned. For the strongest results (CIFAR-100 transfer +7.01%, CIFAR-10 MLP +5.46%), the margins are large enough that seed variance is unlikely to flip the result. For IMDB (+0.39%), it might. We acknowledge this and present the result as suggestive rather than conclusive for that experiment.

5. **No direct NAS comparison.** We compare against random sparsity and dense baselines but not against NAS-discovered architectures. NDNA's advantage over NAS would be in compute cost and transferability, not necessarily in peak accuracy.

6. **Joint optimization.** The genome is trained jointly with network weights, unlike biological evolution where the genome evolves across generations while the brain develops within a lifetime. Separating these timescales could yield different dynamics.

6. Broader Impact and Future Directions

6.1 Scaling to Large Models

If compression ratio continues growing with network size, NDNA could provide significant efficiency gains for large language models. A 258-parameter genome controlling a billion-parameter model's topology would represent extreme compression. The key question is whether the genome's type-based compatibility mechanism can express the wiring patterns needed at that scale.

6.2 Cross-Modality Transfer

We demonstrated transfer within vision (CIFAR-10 to CIFAR-100) and within the same modality (Fashion-MNIST to MNIST). Testing whether a genome trained on vision tasks transfers to language or audio tasks would probe whether the structural principles are truly universal.

6.3 Multi-Genome Evolution

The current approach trains a single genome. A population of genomes competing under selection pressure, analogous to biological evolution, could explore the topology space more broadly and avoid local optima. Each generation would train network weights from scratch using the genome's topology, with genome fitness determined by network performance.

6.4 Adaptive Architectures

A single genome grows a fixed topology. An extension could condition the genome on external signals (hardware constraints, latency budget, task difficulty), producing different topologies for different deployment contexts from the same compact specification.

6.5 Efficient Video Generation

The video experiment demonstrates that NDNA learns structured spatiotemporal sparsity that dramatically outperforms random sparsity (21.7% relative MSE improvement). Video generation models like Sora (Brooks et al., 2024) process millions of spatiotemporal tokens with quadratic attention cost. If NDNA's factored spatiotemporal genome can identify that 89% of attention connections are unnecessary, as demonstrated on Moving MNIST, the compute savings for large-scale video generation could be substantial. The factored genome architecture (separate temporal and spatial sub-genomes) scales naturally: a genome of 5,000-10,000 parameters could potentially control billions of connections in a full-scale video model. Scaling this to natural video with higher resolution and longer sequences is a direct next step.

7. Conclusion

We introduced NDNA, a compact genome of fewer than 400 parameters that learns to grow neural network topology through type-based compatibility rules, default-disconnected initialization, and metabolic cost pressure. Across four architectures (MLP, CNN, Transformer, Video Transformer) and six datasets, NDNA consistently outperforms random sparsity at matched density by 0.39% to 21.7%, matches or exceeds dense baselines, and transfers topology across tasks without modification.

The genome achieves compression ratios from 88:1 to 8,384:1, scaling favorably with network size. On IMDB, 258 genome parameters discover a transformer topology that beats the standard dense architecture. On CIFAR-100, a genome frozen from CIFAR-10 training outperforms random sparse wiring by 7.01 percentage points. On Moving MNIST, a factored spatiotemporal genome of 374 parameters matches the dense video transformer within 0.1% while random sparsity at the same density performs 21.7% worse, the largest gap in any experiment. The genome independently discovered temporal recency and spatial locality, structural principles that mirror hand-designed video processing algorithms.

These results suggest that useful neural network architectures occupy a low-dimensional manifold navigable by simple developmental programs. A compact set of growth rules, not a detailed wiring diagram, suffices to specify effective network topology. The factored spatiotemporal extension demonstrates that NDNA's developmental framework extends naturally to structured domains beyond static data. This perspective opens the door to learned architecture generation that is cheap, transferable, and scalable.

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Appendix A: Failed Approaches

The final NDNA design emerged after seven phases of experimentation spanning approximately 60 hours of compute. We document these failures because they shaped the final design and because negative results are informative.

Phase 1: Weight Sharing (Hours 1 to 8). A base weight matrix was shared across layers with per-layer low-rank corrections, inspired by DNA shared across cell types. Result: 98.8% of normal quality with 41% fewer parameters. However, statistical validation showed the advantage was indistinguishable from dropout regularization. The genome added no structural insight beyond noise injection.

Phase 2: Weight Generation (Hour 9). A tiny neural network (the "genome") generated weight values for the entire model, inspired by how genes encode proteins. Result: 25% worse than baseline. Factored template approaches also failed (27% worse). Key learning: generating weight VALUES is the wrong abstraction. DNA does not specify synapse strengths; it specifies growth rules.

Phase 3: Better Seed Architecture (Hours 10 to 10b). Per-type weight ranks, gated corrections, and split learning rates. Best result: 1.46% worse than baseline. The seed's advantage was again indistinguishable from regularization.

Phase 4: Developmental Topology Without Default-Disconnected (Hours 11 to 11d). Cell types and growth rules produced sparse connections, but initialization was neutral (not negative). Result: 98.0% accuracy (vs 98.6% normal) with 3x fewer parameters. But when parameter-matched, the normal dense MLP won by 0.17% to 0.30%. The genome produced topology, but not useful topology.

Phase 5: Growing Networks (Hours 12 to 12b). Networks that grow during training by adding neurons and splitting like cell division. Result: grew from 16 to 56 neurons reaching 95.9%, but a standard MLP with 56 neurons achieves 97.8%. Growing from small is worse than starting at the right size.

Phase 6: Learned Learning Rules (Hour 13). The genome specified local Hebbian-like learning rules per cell type, removing backpropagation. Result: 11% to 19% accuracy on MNIST (random chance is 10%). Complete failure. Local learning without global error signal cannot solve even simple tasks.

Phase 7: Default-Disconnected Breakthrough (Hours 14 to 14b). The same type-based topology mechanism from Phase 4, but with compatibility matrix initialized negative (default disconnected) and metabolic cost pressure. Result: 97.54% at 2.2% density, beating random sparse by 8.02 percentage points on the original MNIST experiment. Skip connections emerged without being designed. The genome learned general wiring principles. Everything before this was searching for the right inductive bias; default-disconnected was it.

Appendix B: Transformer Implementation Note

The genome transformer requires manual multi-head attention rather than PyTorch's `nn.MultiheadAttention`. The standard module leaves the attention output path unmasked, giving the genome no reason to grow attention connections (they flow freely), so it converges to 0% density and contributes nothing. Replacing it with manual multi-head attention and masking the output projection W_O ensures every information path passes through a genome gate, forcing the genome to find useful wiring. This is a prerequisite for the genome to function in transformer architectures.

Appendix C: Hyperparameters

Table C1: Genome Hyperparameters (constant across experiments)

Parameter	Value
Number of types (K)	8
Type dimension (D)	8
Affinity initialization	$\mathcal{N}(0, 0.1^2)$
Compatibility initialization	$\mathcal{N}(-1.0, 0.3^2)$
Connection scale initialization	3.0
Depth penalty initialization	2.0
Band type base initialization	$\mathcal{N}(0, 0.5^2)$
Band type gradient initialization	$\mathcal{N}(0, 0.3^2)$
Type softmax temperature	3.0
Random seed	42

Table C2: Training Hyperparameters Per Experiment

Experiment	Bands	Epochs	Weight Optimizer	Weight LR	Genome LR	λ	Temp Ra
MNIST MLP	6	25	Adam	1e-3	1e-3	0.1	1.0 (fixed)
CIFAR-10 MLP	6	40	Adam	1e-3	1e-3	0.1	1.0 (fixed)
CIFAR-10 CNN	8	200	SGD (m=0.9, wd=1e-4)	0.1	1e-3	0.01	1.0 to 10.0
CIFAR-100 Transfer	8	250	SGD (m=0.9, wd=5e-4)	0.1	1e-3	0.01	1.0 to 10.0
IMDB Transformer	8	15	AdamW (wd=0.01)	2e-4	0.01	0.005	1.0 to 10.0
Moving MNIST Video	6 (depth) + factored	50	AdamW (wd=0.01)	3e-4	0.01 (depth), 0.01 (attn)	0.005	1.0 to 10.0

Note: MLP experiments use a single Adam optimizer for both genome and weight parameters. CNN and Transformer experiments use split optimizers. All experiments use cosine annealing learning rate schedules. Batch sizes: 128 for vision tasks, 16 for IMDB (with gradient accumulation of 2 steps), 32 for

video (with gradient accumulation of 2 steps). The video experiment uses a factored genome with separate temporal (74 params, 4 types), spatial (74 params, 4 types), and depth (226 params, 8 types) subgenomes with three independent optimizers.